1. Large-prime variation in the cover attack: lead L6 of #2216 (2026-10-18)

Verdict (derived, with measured constants). For the cover decomposition path of `examples/jv_cover.rs`, a large-prime variation lowers `S` by at most 0.02 % (one large prime), 0.27 % (two) and 0.56 % (an information bound no scheme is known to reach) for every p ≤ 12000. The 10 % gate fails by more than a factor 18. No attack code was changed; the study adds opt-in logging to the dlp path and a model example.

Labels: **measured** = a recorded run; **derived** = computed from measured values (formula in `COSTMODEL.md`); **proposed** = not built. No wall-clock claims; no new cross-method comparison with rho is made, so the §5a price band is not needed (rho figures quoted from other studies are lower-bound comparisons as recorded there).

1.1. Boundary, target, falsification

- **Boundary (derived).** Any elimination of the points outside the sub-base `F'` needs at least `touched + 1` solutions (generic coefficients), `c* N` with `c* = 0.9975`, whatever `|F'|` is. The plain method uses `N + 1`.
- **Target set before measuring.** Model saving ≥ 10 % of `S_full` for some p ≤ 12k; otherwise write the negative result.
- **Class.** Negative result; no row improves on the baseline.

1.2. Cost model

`S(f) = X(f) · T · C_test + 419 (f N)^2`, `N = |F| ≈ p/2`, `f = |F'|/|F|`, `α = log|F'|/log|F|`.

- `T` (tests per decomposition), measured 779 ± 33 (95 %; 14 runs, 2139 decompositions; the heuristic is 6! = 720).
- `C_test`, measured 1.2216M counted multiplications at p = 101 (#2182 post-#1717 records), growing 0.75 % per doubling of p (pre-#1717 ratio, derived).
- LA, measured 26.2 Wiedemann operations × 16 multiplications per `u^2` at row weight 7.
- `X(f)/N`: solutions needed, Monte Carlo over uniform 6-subsets of `F` (`curves.json`).
- A decomposition test returns decompositions over the whole `F` at a cost independent of `F'` (argued): the sub-base only filters afterwards. Yields: full `f^6`, 1-partial `6 f^5 (1-f)`, 2-partial `15 f^4 (1-f)^2`.
- Graph cost priced 0, merged-row weight ignored: both favour large primes.

1.3. Measured against modelled yields

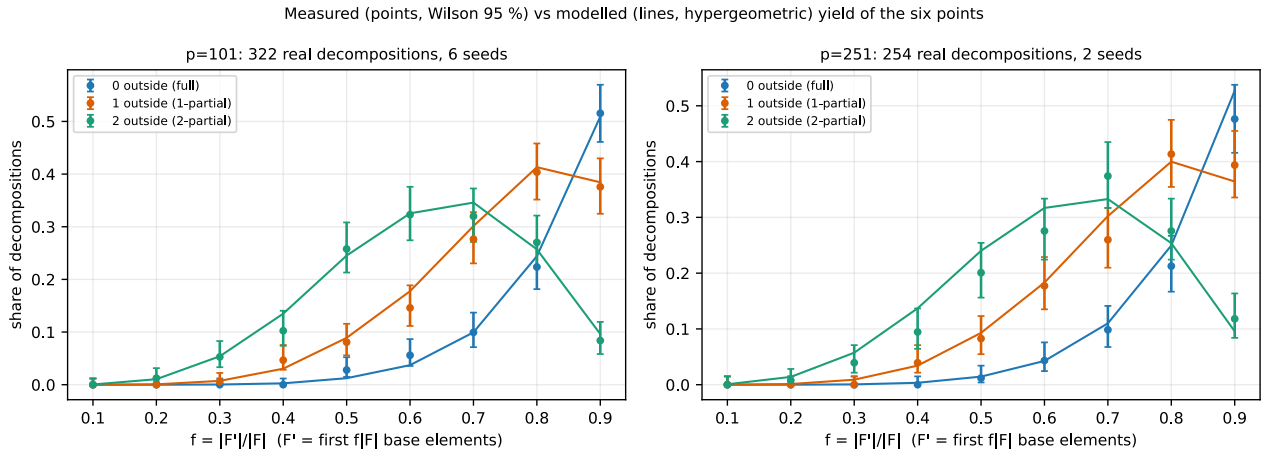


Figure 1: Share of real decompositions with 0, 1, 2 points outside a sub-base (first f|F| base elements). Points: measured with Wilson 95 % intervals (322 decompositions at p = 101, 6 seeds; 254 at p = 251 (2 seeds), 1 seed). Lines: hypergeometric model. Records `raw/yields.json`, logs `raw/dlp_p*_s*.log`.

The six abscissae are uniform (chi-square z between −1.5 and +0.6 per seed; all six distinct). The largest |z| over 72 cells is 2.6. Real decompositions give each scheme the same number of relations as the uniform model (p = 101, f = 0.5: single prime 1.10 real against 1.07 modelled; bound 27.67 against 27.71). Controls: seeds 1, 2 at p = 101 and seeds 1, 2 at p = 251 reproduce the earlier records' residual counts and multiplication totals; every discrete logarithm verified (`correct = true`).

1.4. Solutions needed

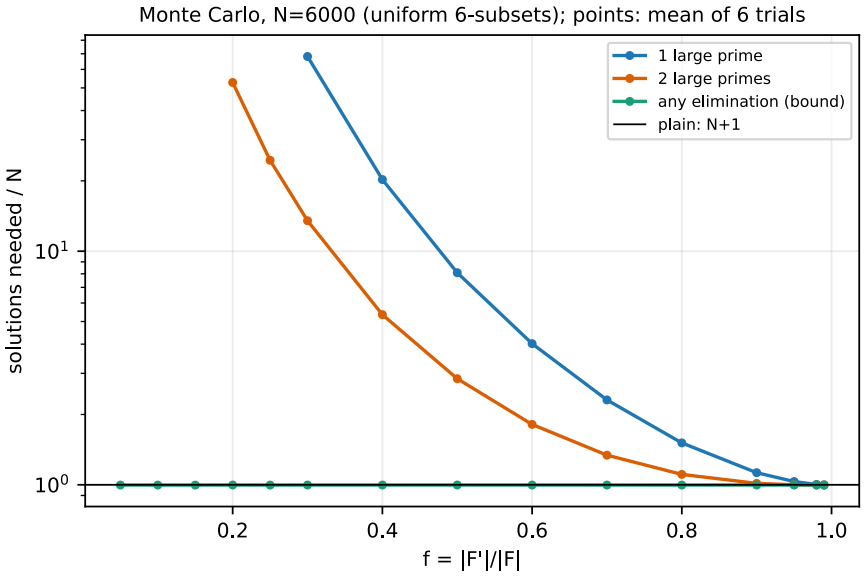


Figure 2: Monte Carlo, N = 6000: solutions needed over N for one large prime, two, and the information bound, against f. The plain method needs N + 1. Mean of the trials in `raw/curves.json` (N-independence checked from 484 to 6000 within 3 %).

1.5. Result

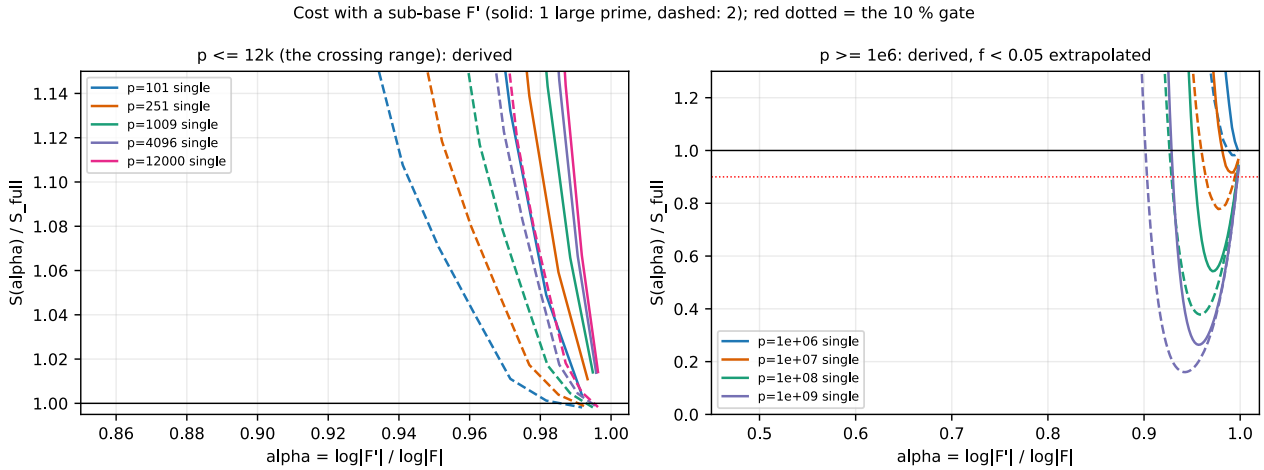


Figure 3: Derived S(α)/S_full for one (solid) and two (dashed) large primes. Left: p ≤ 12k, where no curve goes below 0.997. Right: p ≥ 1e6, where the LA is large enough to trade (single-prime and double-prime rows use the N = 6000 curve; extrapolated below f = 0.05 for the bound only). `raw/cost.json`.

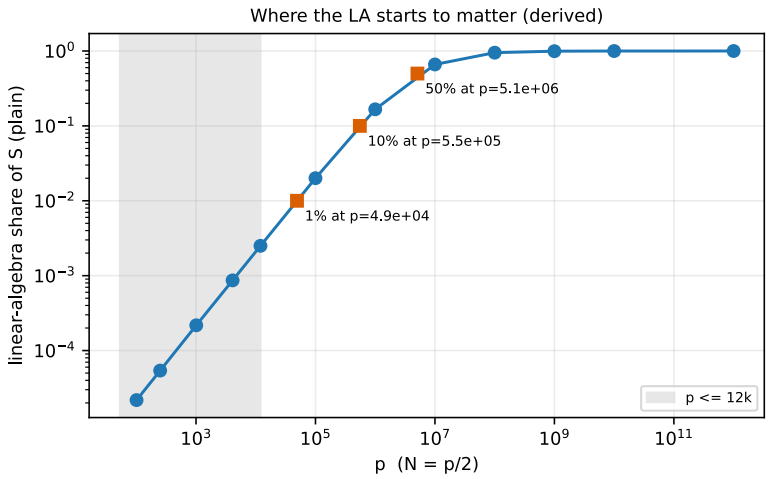


Figure 4: Derived LA share of the plain method, `419 N^2 / ((N+1) T C_test + 419 N^2)`, N = p/2. 1 % at p ≈ 4.9e4, 10 % at 5.5e5, 50 % at 5.1e6.

p	LA share	1 large prime	2 large primes	bound
101	2.2e-5	0.9995	0.9981	0.9980
251	5.4e-5	0.9991	0.9978	0.9944
1009	2.2e-4	0.9997	0.9973	0.9966
4096	8.7e-4	0.9999	0.9973	0.9957
12000	2.5e-3	0.9998	0.9975	0.9944
1e6	0.166	0.996	0.980	0.832
1e7	0.661	0.916	0.778	0.34
1e8	0.950	0.542	0.378	0.050
1e12	1.000	0.029	0.013	7e-6

Best S/S_full over f (derived). Below 12k the differences from 1 are at the Monte-Carlo noise level. The 10 % gate is passed at p between 1e6 and 1e7 (two large primes) and between 1e7 and 1e8 (one), far above the crossing range. Asymptotically a single large prime gives `S ∝ N^(2 - 4/13)` against `N^2` (derived): the exponent changes, but only where the LA already dominates.

1.6. What would change the exponent, ranked

1. The sieve path (`jv_sieve.rs`) with a large prime (proposed): LA share there is measured 8.5–9.9 % at p = 1823 (0.04–0.07 % at 251), relation cost 2.1–2.7M multiplications at p = 1823 against about 0.95G for 779 cover tests (derived).
2. A different m: free in the sieve (11 at p = 101, 9 from 1009), fixed at 6 in the decomposition path.
3. A cheaper test: constant factor, L1–L4 of #2216 reach about 1.4×.
4. Large primes in the decomposition path: at most 0.56 % for p ≤ 12k.

1.7. Correctness

`cargo test --release --lib jv_cover`: see `raw/test_jv_cover.log`. Lean (`lean/LpBound.lean`, toolchain v4.34.1, axioms `propext`, `Quot.sound`): the conditional arithmetic step only ("if a sub-base scheme needs at least the plain solutions, it saves at most the plain LA cost"). The rank bound itself is open (needs linear algebra, no Mathlib).

1.8. Open, limits

Proposition 1 (no F'-restricted test is cheaper) is argued, not proved (`problems/jv-cover-subbase-restricted-test.md`). Coefficients are generic in the bound. Monte Carlo N ≤ 6000. Two seeds at p = 251. The sieve with large primes is not modelled here.

1.9. Sources

#2216 `research/ic_cover_leads_20261018/README.md`; #2182 `research/f4_fp_ic_effect_20261018/` (records `raw/ref/pr2182_*.json`); `experiments/30_jv_cover_dlp*.json`, `30_jv_cover_ccov.json`, `32_*`, `35_*`, `38_*`; Gaudry–Thomé–Thériault–Diem, Math. Comp. 76 (2007); Joux–Vitse, Eurocrypt 2012.